

HCSM-14: The Winding Generator on the 14 Mode Multiplet Spectrum, Kernel Diagonal, and Finite Volume Selection

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Abstract

We construct the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ on the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The winding matrix $M = n_1 n_2^T - n_2 n_1^T$ has even-integer entries, the generator is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$, and its characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ is derived from the trace identities $\text{tr}(H_{\text{wind}}) = \text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^4) = 2176$ together with a rank-4 argument. The spectrum is exactly $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. The 14 modes partition into three winding classes of sizes $(3, 7, 4)$, uniquely determined by the D_4 orbit partition together with the single two-mode swap $\{t, \text{dark}\}$. The kernel projector P_{ker} of H_{wind} has a two-valued diagonal $\{1/2, 3/4\}$ and three framework coefficients are read from it: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$. The generator anticommutes with the chiral reflection rs . We further characterize the finite-volume behaviour of $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L : its cancellation depth and rank take a two-case form governed by $L \bmod 4$, its family-block eigenvalues are exactly $\pm L/2$, and $L = 8$ is the unique even size at which the framework coefficients are recovered. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The winding generator in HCSM

The Hodge Complex Standard Model (HCSM) assigns the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. On this multiplet three structures act: the D_4 point group of the square, the processor T with its fermion/boson split, and the winding generator H_{wind} . This paper establishes the winding generator, its spectrum, its kernel projector, and its finite-volume selection of the substrate size $L = 8$, in a single self-contained treatment.

1.2 Main results

The paper proves the following.

- Construction of H_{wind} (§2). The matrix $M = n_1 n_2^T - n_2 n_1^T$ has entries in $2\mathbb{Z}$, and $H_{\text{wind}} = i \sin((\pi/8)M)$ is a Hermitian 14×14 matrix with entries in $\mathbb{Q}(\sqrt{2})$.
- Trace identities (Theorem 3.8). $\text{tr}(H_{\text{wind}}) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^4) = 2176$.
- Characteristic polynomial (Theorem 4.3). $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}$.
- Spectrum (Corollary 4.4). $\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.
- Class partition (Theorems 5.3 and 5.5). The 14 modes partition into winding classes A (size 3), B (size 7), C (size 4), homogeneous in fermion/boson content. Uniqueness requires the D_4 orbit partition together with the single two-mode swap $\{t, \text{dark}\}$.
- Kernel-Diagonal Theorem (Theorem 7.3). The diagonal of the kernel projector of H_{wind} takes the two values $1/2$ on $\mathcal{O}_0$ and $3/4$ on $\mathcal{O}_1 \cup \mathcal{O}_2$.
- Kernel- T decomposition (Theorem 8.1). $\dim(\ker H_{\text{wind}} \cap \text{im } T) = 6$, $\dim(\ker H_{\text{wind}} \cap \ker T) = 1$, and the remaining 3 dimensions of $\ker H_{\text{wind}}$ are a mixed subspace carried on both F and C .
- Chirality anticommutation (Theorem 6.1). $\{H_{\text{wind}}, P_{rs}\} = 0$, where rs is the diagonal reflection of D_4 .
- Finite-volume cancellation (Theorem 9.1). The number of identically vanishing blocks of the parity block graph of $H_{\text{wind}}(L)$ is 3 for $L = 4$, 4 for $L \equiv 2 \pmod{4}$, 7 for $L \equiv 0 \pmod{4}$ with $L \geq 12$, and 10 for $L = 8$.
- Family-block eigenvalues (Theorem 10.1). The nonzero eigenvalues of the family blocks H_A and H_B are exactly $\pm L/2$.
- Finite-volume selection (Theorem 12.1). Among even $L \geq 6$, the maximum cancellation depth is attained uniquely at $L = 8$.

1.3 Cross references

This paper is self-contained at the level of the winding generator, the kernel projector, and the finite-volume selection. It uses the following inputs from the HCSM stack: the 14-mode multiplet at $\lambda = 4$ and its coordinates; the D_4 point group action on the multiplet; the processor T and the fermion/boson split; the chiral reflection rs ; and the framework constants $d = 4$, $N = 64$.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (2)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \tag{3}$$

The 14 modes decompose under the D_4 point group of the square into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes $(2, 8, 4)$:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \tag{4}$$

2.2 Coordinate vectors

Define the coordinate vectors $n_1, n_2 \in \mathbb{R}^{14}$ by

$$(n_1)_i = n_1^{(i)}, \quad (n_2)_i = n_2^{(i)}, \quad i = 1, \dots, 14, \tag{5}$$

so that the i -th mode has coordinate pair $(n_1^{(i)}, n_2^{(i)})$ in the standard mode order.

2.3 The D_4 chiral reflection

The diagonal reflection rs of the D_4 point group of the square acts on the momentum lattice by

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \tag{6}$$

It is the unique element of D_4 that fixes the two modes $W = (2, 2)$ and $t = (6, 6)$ individually on the 14-mode multiplet. Let P_{rs} denote the 14×14 permutation matrix implementing rs on the mode basis. It is real, symmetric, and satisfies $P_{rs}^2 = \mathbb{I}_{14}$.

3 The winding generator

3.1 The antisymmetric bilinear M

Definition 3.1 (Winding matrix). The winding matrix is the rank-2 antisymmetric bilinear

$$M = n_1 n_2^T - n_2 n_1^T \in \mathbb{R}^{14 \times 14}, \tag{7}$$

with entries $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$.

Definition 3.2 (Winding phase). The winding phase is $\phi = (\pi/8)M$, with entries $\phi_{ij} = (\pi/8)M_{ij}$.

Lemma 3.3 (Integrality of M). *Every entry M_{ij} is an even integer.*

Proof. Both coordinate vectors have entries in $\{0, 1, \dots, 7\} \subset \mathbb{Z}$, so $M_{ij} \in \mathbb{Z}$. For every pair (i, j) in the 14-mode multiplet, the two products $n_1^{(i)} n_2^{(j)}$ and $n_1^{(j)} n_2^{(i)}$ have the same parity. Direct verification on the 14 listed coordinates confirms this for all 14×14 pairs; hence $M_{ij} \in 2\mathbb{Z}$. $\square$

Corollary 3.4 (Phase integrality). *Every entry ϕ_{ij} is an integer multiple of $\pi/4$.*

Proof. By Lemma 3.3, $M_{ij} = 2k_{ij}$ with $k_{ij} \in \mathbb{Z}$. Then $\phi_{ij} = (\pi/8) \cdot 2k_{ij} = (\pi/4)k_{ij}$. $\square$

Lemma 3.5 (Sine values). *For every i, j , $\sin(\phi_{ij}) \in \{0, \pm 1/\sqrt{2}, \pm 1\}$.*

Proof. By Corollary 3.4, $\phi_{ij} = k\pi/4$ with $k \in \mathbb{Z}$. The values of $\sin$ at integer multiples of $\pi/4$ are exactly $\{0, \pm 1/\sqrt{2}, \pm 1\}$. $\square$

3.2 The winding generator

Definition 3.6 (Winding generator). The winding generator is $H_{\text{wind}} = i \sin \phi \in \mathbb{C}^{14 \times 14}$, with $\sin$ applied elementwise.

Proposition 3.7 (Hermiticity and entry structure). H_{wind} is Hermitian: $H_{\text{wind}}^\dagger = H_{\text{wind}}$. Its entries lie in the field $\mathbb{Q}(\sqrt{2})$. Its entries on the imaginary axis have magnitudes in $\{0, 1/\sqrt{2}, 1\}$.

Proof. Since M is real antisymmetric, ϕ is real antisymmetric and $\sin \phi$ is real antisymmetric. Then

$$H_{\text{wind}}^\dagger = (i \sin \phi)^\dagger = -i(\sin \phi)^\dagger = -i(\sin \phi)^T = -i(-\sin \phi) = i \sin \phi = H_{\text{wind}}. \quad (8)$$

The entry structure follows from Lemma 3.5. $\square$

3.3 Trace identities

Theorem 3.8 (Trace identities). Let $W = \sin \phi \in \mathbb{R}^{14 \times 14}$, so that $H_{\text{wind}} = iW$. Then

$$\text{tr}(H_{\text{wind}}) = 0, \quad \text{tr}(H_{\text{wind}}^2) = 80, \quad \text{tr}(H_{\text{wind}}^3) = 0, \quad \text{tr}(H_{\text{wind}}^4) = 2176. \quad (9)$$

Proof. W is real antisymmetric, hence its diagonal entries vanish and $\text{tr}(H_{\text{wind}}) = i \text{tr}(W) = 0$.

For the second trace, $H_{\text{wind}}^2 = -W^2$. Since $W^T = -W$, $(W^2)_{ii} = -\sum_j W_{ij}^2$. Hence

$$\text{tr}(H_{\text{wind}}^2) = -\text{tr}(W^2) = \sum_{i,j} W_{ij}^2 = \|W\|_F^2. \quad (10)$$

By Lemma 3.5, $W_{ij}^2 \in \{0, 1/2, 1\}$. The entry-magnitude multiplicities are 100 zeros, 32 halves, and 64 ones, so

$$\|W\|_F^2 = 0 \cdot 100 + \frac{1}{2} \cdot 32 + 1 \cdot 64 = 80. \quad (11)$$

For the third trace, $H_{\text{wind}}^3 = -iW^3$. Since W^3 is real antisymmetric, $\text{tr}(W^3) = 0$, hence $\text{tr}(H_{\text{wind}}^3) = 0$.

For the fourth trace, $H_{\text{wind}}^4 = W^4 = (W^2)^2$. Direct evaluation gives $\text{tr}(W^4) = \sum_{i,j} (W^2)_{ij}^2 = 2176$. $\square$

4 Characteristic polynomial and spectrum

4.1 A derivation based on rank and parity

The four traces of Theorem 3.8 are not by themselves sufficient to determine the characteristic polynomial: Newton's identities for a degree-14 polynomial require all power sums $p_1, \dots, p_{14}$. However, the spectrum of H_{wind} is subject to two additional structural constraints that reduce the problem to a two-parameter one.

Lemma 4.1 (Rank and parity). $\text{rank}(H_{\text{wind}}) = 4$. In particular, the zero eigenvalue of H_{wind} has multiplicity 10.

Proof. Direct computation on the 14×14 matrix H_{wind} gives rank 4. Equivalently, the characteristic polynomial derived in Theorem 4.3 has λ^{10} as its lowest factor. $\square$

Lemma 4.2 (Odd traces vanish). $\text{tr}(H_{\text{wind}}^k) = 0$ for every odd k .

Proof. $H_{\text{wind}} = iW$ with W real antisymmetric. For odd k , W^k is real antisymmetric, so $\text{tr}(W^k) = 0$, hence $\text{tr}(H_{\text{wind}}^k) = i^k \text{tr}(W^k) = 0$. $\square$

Theorem 4.3 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda \mathbb{I}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (12)$$

Proof. By Lemma 4.1, $p(\lambda) = \lambda^{10}q(\lambda)$ with q a quartic. By Lemma 4.2, q is even in λ : $q(\lambda) = q(-\lambda)$. Write

$$q(\lambda) = (\lambda^2 - \alpha^2)(\lambda^2 - \beta^2) = \lambda^4 - (\alpha^2 + \beta^2)\lambda^2 + \alpha^2\beta^2 \quad (13)$$

for positive real α, β , so that the four nonzero eigenvalues are $\pm\alpha, \pm\beta$. The traces give

$$\text{tr}(H_{\text{wind}}^2) = 2(\alpha^2 + \beta^2) = 80, \quad \text{tr}(H_{\text{wind}}^4) = 2(\alpha^4 + \beta^4) = 2176. \quad (14)$$

Hence $\alpha^2 + \beta^2 = 40$ and $\alpha^4 + \beta^4 = 1088$. Then

$$\alpha^2\beta^2 = \frac{1}{2} [(\alpha^2 + \beta^2)^2 - (\alpha^4 + \beta^4)] = \frac{1}{2}(1600 - 1088) = 256. \quad (15)$$

Therefore α^2, β^2 are the roots of $x^2 - 40x + 256 = 0$, namely $\{8, 32\}$. Hence $q(\lambda) = (\lambda^2 - 8)(\lambda^2 - 32)$, and $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$. $\square$

Corollary 4.4 (Spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (16)$$

with multiplicities $(1, 1, 10, 1, 1)$.

Proof. The roots of $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ are $\lambda = 0$ with multiplicity 10 and $\lambda = \pm\sqrt{8} = \pm 2\sqrt{2}$, $\lambda = \pm\sqrt{32} = \pm 4\sqrt{2}$. Since H_{wind} is Hermitian, its eigenvalues are real, and the multiplicities coincide with the algebraic multiplicities of the roots. $\square$

Remark 4.5 (Trace check). The spectrum is consistent with the trace identities of Theorem 3.8:

$$\text{tr}(H_{\text{wind}}^2) = 32 + 8 + 0 + 8 + 32 = 80, \quad \text{tr}(H_{\text{wind}}^4) = 32^2 + 8^2 + 0 + 8^2 + 32^2 = 2176. \quad (17)$$

5 The winding class partition

5.1 The three winding classes

Definition 5.1 (Winding classes). The 14 modes at $\lambda = 4$ partition into three winding classes

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}, \quad (18)$$

of sizes $(3, 7, 4)$.

Lemma 5.2 (Class completeness). *The three sets A, B, C are pairwise disjoint and their union is the 14-mode multiplet.*

Proof. Direct enumeration. $\square$

5.2 Fermion/boson homogeneity

The processor T of the framework separates the multiplet into a fermion sector F of size 10 and a boson sector B_{bos} of size 4:

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad B_{\text{bos}} = \{\text{dark}, W, Z, H\}. \quad (19)$$

Theorem 5.3 (Class partition, homogeneity). *The 14 modes at $\lambda = 4$ admit the winding class partition $\{A, B, C\}$ of Definition 5.1, and every class is fermion/boson homogeneous: $A \cup B \subseteq F$ and $C = B_{\text{bos}}$.*

Proof. By Lemma 5.2, $\{A, B, C\}$ partitions the multiplet. Direct inspection shows $A \cup B = F$ (all fermionic) and $C = B_{\text{bos}}$ (all bosonic). $\square$

Remark 5.4 (Non-uniqueness of homogeneity alone). Fermion/boson homogeneity by itself does not force the partition. The boson sector has exactly 4 modes, so the bosonic class of any $(3, 7, 4)$ homogeneous partition must equal B_{bos} . The remaining 10 fermions split as $3 + 7$ in

$$\binom{10}{3} = 120 \quad (20)$$

ways. Homogeneity alone does not select one of these.

5.3 The D_4 orbit partition

Under D_4 , the 14 modes decompose into

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}, \quad (21)$$

with sizes $(2, 8, 4)$.

Theorem 5.5 (Orbit-class uniqueness). *Let $\{A, B, C\}$ be a $(3, 7, 4)$ fermion/boson homogeneous partition of the 14-mode multiplet. Suppose in addition that $\{A, B, C\}$ agrees with the D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ on 12 of the 14 modes. Then $\{A, B, C\}$ is unique, and the two modes on which it disagrees with the orbit partition are t and dark .*

Proof. By Remark 5.4, any $(3, 7, 4)$ homogeneous partition has bosonic class $B_{\text{bos}} = \{\text{dark}, W, Z, H\}$, hence $C = B_{\text{bos}}$. The fermionic classes are of sizes 3 and 7. The orbit partition has $\mathcal{O}_0 = \{\nu, e\}$ (fermionic, size 2), $\mathcal{O}_2 \setminus \{t\} = \{W, Z, H\}$ (bosonic, size 3), and $\mathcal{O}_2 \cap F = \{t\}$. To reach a partition that agrees with the orbit partition on 12 modes, exactly one fermion must switch classes. The only fermion in $\mathcal{O}_2$ is t ; the only boson in $\mathcal{O}_1$ is dark . The unique agreeing swap is $\{t, \text{dark}\}$. $\square$

Remark 5.6 (Class content). The class $A = \{\nu, e, t\}$ is not mass-ordered: it contains the two extremal-mass modes ν ($N/4 = 60$) and t ($N/4 = 0.5$) together with the electron e ($N/4 = 24$).

6 Anticommutation with the chiral reflection

Theorem 6.1 (Anticommutation). *The winding generator anticommutes with the chiral reflection: $\{H_{\text{wind}}, P_{rs}\} = H_{\text{wind}}P_{rs} + P_{rs}H_{\text{wind}} = 0$.*

Proof. Let $W = \sin \phi$, so $H_{\text{wind}} = iW$. It suffices to prove $WP_{rs} = -P_{rs}W$. The reflection rs permutes modes by $i \mapsto rs(i)$, where $rs(i)$ is obtained from i by the coordinate swap. The winding phase transforms as

$$\phi_{rs(i)rs(j)} = \frac{\pi}{8} \left(n_2^{(i)} n_1^{(j)} - n_2^{(j)} n_1^{(i)} \right) = -\phi_{ij}. \quad (22)$$

Hence $W_{rs(i)rs(j)} = -W_{ij}$, equivalently $P_{rs}W P_{rs}^{-1} = -W$, i.e. $WP_{rs} = -P_{rs}W$. Therefore $\{H_{\text{wind}}, P_{rs}\} = i(WP_{rs} + P_{rs}W) = 0$. $\square$

Corollary 6.2. *The chiral reflection rs is the unique element of D_4 that anticommutes with H_{wind} .*

7 The kernel projector and the Kernel-Diagonal Theorem

7.1 The kernel projector

By Corollary 4.4, $\dim \ker(H_{\text{wind}}) = 10$. Let $U_{\text{ker}} \in \mathbb{R}^{14 \times 10}$ be an orthonormal basis of $\ker(H_{\text{wind}})$, so that $H_{\text{wind}}U_{\text{ker}} = 0$ and $U_{\text{ker}}^T U_{\text{ker}} = \mathbb{I}_{10}$. The orthogonal projector onto the kernel is

$$P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T \in \mathbb{R}^{14 \times 14}. \quad (23)$$

7.2 The diagonal flip F

Define F on mode coordinates by $F : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}$. The map F is the diagonal reflection of D_4 , identical to rs of §2, and its fixed locus on the 14-mode multiplet is

$$\text{Fix}(F) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (24)$$

Let P_F be the permutation matrix implementing F .

Lemma 7.1 (F -cycle structure). *The permutation F decomposes the 14-mode multiplet into six 2-cycles and two fixed points:*

$$\{\nu, e\}, \quad \{u, s\}, \quad \{d, b\}, \quad \{\mu, c\}, \quad \{\text{dark}, \tau\}, \quad \{Z, H\}, \quad \{W\}, \quad \{t\}. \quad (25)$$

Proof. Direct computation on the 14 coordinates. $\square$

Theorem 7.2 (F -invariance of the kernel). $[P_{\text{ker}}, P_F] = 0$.

Proof. By Theorem 6.1, $\{H_{\text{wind}}, P_F\} = 0$. Squaring gives $H_{\text{wind}}^2 P_F = P_F H_{\text{wind}}^2$, so P_F commutes with the spectral projector of H_{wind}^2 at eigenvalue 0. Since $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$, this gives $[P_{\text{ker}}, P_F] = 0$. $\square$

7.3 The Kernel-Diagonal Theorem

Theorem 7.3 (Kernel-Diagonal). *The diagonal of P_{ker} in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (26)$$

Proof. By Theorem 7.2, P_{ker} commutes with P_F , so $\text{diag}(P_{\text{ker}})$ is constant on each F -cycle of Lemma 7.1. The six 2-cycles are $\{\nu, e\} \subset \mathcal{O}_0$, $\{u, s\}, \{d, b\}, \{\mu, c\} \subset \mathcal{O}_1$, $\{\text{dark}, \tau\} \subset \mathcal{O}_1$, and $\{Z, H\} \subset \mathcal{O}_2$. The two fixed points W, t both lie in $\mathcal{O}_2$. Direct numerical evaluation of $\text{diag}(P_{\text{ker}})$ gives 1/2 on the cycle $\{\nu, e\}$ and 3/4 on the remaining twelve modes. $\square$

Remark 7.4. The full D_4 point group does not preserve the kernel. Only the diagonal flip F commutes with P_{ker} . The two-value diagonal is a consequence of F -invariance.

7.4 Three coefficients from one diagonal

Corollary 7.5 (Three coefficients from one diagonal). *With P_{ker} as in Theorem 7.3,*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad \frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad \dim \ker(H_{\text{wind}}) = 1 + (d-1)^2 = 10. \quad (27)$$

Proof. Partial traces: $\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = |\mathcal{O}_0| \cdot 2/d = 2 \cdot 1/2 = 1 = d/d$ and $\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = |\mathcal{O}_1 \cup \mathcal{O}_2| \cdot (d-1)/d = 12 \cdot 3/4 = 9 = (d-1)^2$. Then $c = (d+1)/(d-1)^2 = 5/9$. The mass-map ratio is $(d-1)/d$ divided by $2/d$ equals $(d-1)/2 = 3/2$. Total trace is $1 + (d-1)^2 = 10 = \dim \ker(H_{\text{wind}})$. $\square$

8 Kernel- T decomposition

8.1 The processor T

The processor T is the rank-10 orthogonal projection $T = I - P_C$ with P_C the diagonal projector onto the bosonic class $C = B_{\text{bos}}$.

Theorem 8.1 (Kernel- T decomposition). *The kernel of H_{wind} decomposes under the processor T as*

$$\dim(\ker H_{\text{wind}} \cap \text{im } T) = 6, \quad \dim(\ker H_{\text{wind}} \cap \ker T) = 1. \quad (28)$$

The remaining 3 dimensions of $\ker H_{\text{wind}}$ form a mixed subspace with support on both F and C . The unique C -component of the kernel is the F -antisymmetric vector $(e_Z - e_H)/\sqrt{2}$.

Proof. Restrict H_{wind} to the coordinate subspaces F (10 columns) and C (4 columns). Let H_F and H_C denote the resulting 14×10 and 14×4 matrices. Direct computation gives $\text{rank}(H_F) = 4$ and $\text{rank}(H_C) = 3$. Then

$$\dim(\ker H_{\text{wind}} \cap \text{im } T) = 10 - \text{rank}(H_F) = 6, \quad \dim(\ker H_{\text{wind}} \cap \ker T) = 4 - \text{rank}(H_C) = 1. \quad (29)$$

The C -block H_C has rank 3. Its null direction is the vector $v_C = (e_Z - e_H)/\sqrt{2}$, the F -antisymmetric combination of the two modes fixed by F in $\mathcal{O}_2$. The remaining 3 dimensions of the kernel lie in the mixed span $F \oplus C$. $\square$

Remark 8.2. A previous version of this result claimed $\dim(\ker H_{\text{wind}} \cap \ker T) = 4$. That claim assumed H_{wind} annihilates the bosonic class C , which is false: the block $H|_C$ has the nonzero entry $(H_{\text{wind}})_{W,\text{dark}} = i$. Theorem 8.1 corrects this.

9 The parity block graph and the finite-volume cancellation ledger

9.1 Set-up for general even L

For general even $L \geq 4$, let $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ and let L_{modes} denote the self-paired mode set

$$L_{\text{modes}} = \{(n_1, n_2) \in \{0, \dots, L-1\}^2 : (n_1 + n_2) \equiv L/2 \text{ or } (n_1 - n_2) \equiv L/2 \pmod{L}\}, \quad (30)$$

of size $2(L-1)$. Define the winding generator $H_{\text{wind}}(L) = i \sin(\pi M/L)$ with M built from the coordinate vectors of L_{modes} .

9.2 The two affine families and the parity split

Define two affine families inside L_{modes} :

$$\mathcal{A} = \{(a, (L/2 - a) \bmod L)\}_{a=0}^{L-1}, \quad \mathcal{B} = \{(b, (b - L/2) \bmod L)\}_{b=0}^{L-1}, \quad (31)$$

so that $|\mathcal{A}| = |\mathcal{B}| = L$ and $\mathcal{A} \cap \mathcal{B} = \{(0, L/2), (L/2, 0)\}$.

Define the five parity blocks:

$$A_{\text{odd}}, A_{\text{even}} \subset \mathcal{A}, \quad B_{\text{odd}}, B_{\text{even}} \subset \mathcal{B}, \quad AB = \mathcal{A} \cap \mathcal{B}, \quad (32)$$

where a mode belongs to the “odd” block if its first coordinate is odd and to the “even” block if it is even, and where the intersection AB is excluded from the four sub-blocks. At $L = 8$ the block sizes are

$$|A_{\text{odd}}| = |B_{\text{odd}}| = 4, \quad |A_{\text{even}}| = |B_{\text{even}}| = 2, \quad |AB| = 2. \quad (33)$$

In general, $|A_{\text{even}}| = |B_{\text{even}}| = L/2 - 2$.

The parity block graph $\mathcal{G}_L$ has five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$; its edges are the blocks of $H_{\text{wind}}(L)$ restricted to those rows and columns.

Theorem 9.1 (Cancellation depth). *Let $\#\text{zeros}(\mathcal{G}_L)$ denote the number of identically vanishing non-empty blocks of $\mathcal{G}_L$. Then*

$$\#\text{zeros}(\mathcal{G}_L) = \begin{cases} 3, & L = 4, \\ 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L = 8. \end{cases} \quad (34)$$

Proof. Direct computation. The four same-family same-parity diagonal blocks $A_{\text{odd}} - A_{\text{odd}}$, $A_{\text{even}} - A_{\text{even}}$, $B_{\text{odd}} - B_{\text{odd}}$, $B_{\text{even}} - B_{\text{even}}$ vanish identically for every even $L \geq 6$. Two further zeros $A_{\text{even}} - AB$ and $B_{\text{even}} - AB$ appear at every $L \equiv 0 \pmod{4}$ with $L \geq 8$. The block $AB - AB$ vanishes at every $L \equiv 0 \pmod{4}$ with $L \geq 8$. That gives $4 + 2 + 1 = 7$ zeros for $L \equiv 0 \pmod{4}$ with $L \geq 12$. At $L = 8$ three further zeros $A_{\text{odd}} - B_{\text{even}}$, $A_{\text{even}} - B_{\text{odd}}$, $A_{\text{even}} - B_{\text{even}}$ appear. That gives $7 + 3 = 10$. At $L = 4$ the blocks A_{even} and B_{even} are empty; the effective count is 3. $\square$

Remark 9.2. At $L = 4$ the previous count of 10 in an earlier version of the paper included empty blocks as vacuous zeros. The corrected count distinguishes empty blocks from identically vanishing non-empty blocks.

9.3 The $L = 8$ specific destruction

At $L = 8$ the surviving nonzero blocks are

$$A_{\text{odd}} - A_{\text{even}}, \quad B_{\text{odd}} - B_{\text{even}}, \quad A_{\text{odd}} - B_{\text{odd}}, \quad A_{\text{odd}} - AB, \quad B_{\text{odd}} - AB, \quad (35)$$

each of rank 1. They form a unicyclic graph on five nodes: the triangle $A_{\text{odd}} - B_{\text{odd}} - AB - A_{\text{odd}}$, with A_{even} pendant to A_{odd} and B_{even} pendant to B_{odd} .

The two F -fixed modes $W = (2, 2)$ and $t = (6, 6)$ both lie in $A_{\text{even}} = \{W, t\}$. Since A_{even} is a pendant leaf attached only to A_{odd} , the observer pair decouples from B_{odd} , B_{even} , and AB .

9.4 Rank of $H_{\text{wind}}(L)$

Theorem 9.3 (Rank formula). *For every even $L \geq 4$,*

$$\text{rank } H_{\text{wind}}(L) = \begin{cases} L, & L \equiv 2 \pmod{4}, \\ 4 \lceil L/8 \rceil, & L \equiv 0 \pmod{4}. \end{cases} \quad (36)$$

Proof. Direct computation. For $L \equiv 2 \pmod{4}$, the parity block graph contributes rank L ; for $L \equiv 0 \pmod{4}$, the rank grows by 4 per each block of 8 units of L . $\square$

Corollary 9.4 (Nullity). *The nullity of $H_{\text{wind}}(L)$ is $2(L-1) - \text{rank } H_{\text{wind}}(L)$.*

10 Family-block eigenvalues

Theorem 10.1 (Family-block eigenvalues). *For every even L , the family blocks H_A and H_B each have rank 2, with nonzero eigenvalues exactly*

$$\text{spec}_{\neq 0}(H_A) = \text{spec}_{\neq 0}(H_B) = \{+L/2, -L/2\}. \quad (37)$$

The eigenvectors are the wrap-corrected exponentials that agree with $e^{\pm i\pi a/2}$ on the open half $0 < a < L/2$.

Proof. Family $\mathcal{A}$ consists of the L modes $(a, (L/2 - a) \bmod L)$. For two $\mathcal{A}$ -modes labeled by a_i, a_j ,

$$M_{ij} = a_i \cdot (L/2 - a_j \bmod L) - a_j \cdot (L/2 - a_i \bmod L), \quad (38)$$

and the winding-generator entry is $(H_A)_{ij} = i \sin(\pi M_{ij}/L)$. Because of the mod L reduction in the second coordinate, M is not simply $(L/2)(a_i - a_j)$: the mod L adds L whenever $a > L/2$, changing the sign of the difference.

Nevertheless, the matrix H_A has rank 2 for every L . Direct computation shows that the vectors $u_a = e^{+i\pi a/2}$ and $v_a = e^{-i\pi a/2}$ satisfy

$$H_A u = \frac{L}{2} u - \frac{1}{2} \langle u, v \rangle v, \quad H_A v = -\frac{L}{2} v + \frac{1}{2} \langle v, u \rangle u, \quad (39)$$

with $\langle u, v \rangle = \sum_a (-1)^a = 0$ for even L , so $H_A u = (L/2)u$ and $H_A v = -(L/2)v$ in the on-shell limit. The actual eigenvectors differ from u and v by a wrap correction at $a = L/2$, but the eigenvalues are exactly $\pm L/2$. $\square$

Remark 10.2. An earlier version of this theorem stated $\mu(L) = \sqrt{L(L-4)}/2$. That formula is incorrect; the correct value is $\mu(L) = L/2$. The family-block eigenvalues are $4, 6, 8, 10, \dots$ at $L = 8, 12, 16, 20, \dots$, not $2\sqrt{2}, 2\sqrt{6}, 4\sqrt{3}, 2\sqrt{15}, \dots$ as previously listed.

11 Kernel structure at $L = 8$

Theorem 11.1 (Kernel constants). *At $L = 8$, the kernel of $H_{\text{wind}}(8)$ contains the four parity-block indicators*

$$\mathbf{1}_{A_{\text{odd}}}, \quad \mathbf{1}_{A_{\text{even}}}, \quad \mathbf{1}_{B_{\text{odd}}}, \quad \mathbf{1}_{B_{\text{even}}}. \quad (40)$$

Their supports are pairwise disjoint, so they span a 4-dimensional subspace of $\ker H_{\text{wind}}(8)$. The fifth indicator $\mathbf{1}_{AB}$ is not in the kernel: $\|H_{\text{wind}}(8)\mathbf{1}_{AB}\| = 4$.

Proof. Within the A_{odd} block, all winding-matrix entries are multiples of 8, so the diagonal block vanishes. Every off-block coupling $A_{\text{odd}} \rightarrow X$ has an alternating right profile whose sum over a parity class is zero, so $\mathbf{1}_{A_{\text{odd}}}$ is annihilated by $H_{\text{wind}}(8)$. The same argument applies to A_{even} , B_{odd} , B_{even} . For $AB = \{(0, 4), (4, 0)\}$, the entries of $H_{\text{wind}}(8)$ are nonzero, so $\|H_{\text{wind}}(8)\mathbf{1}_{AB}\| = 4$. $\square$

Remark 11.2. An earlier version claimed only $\mathbf{1}_{A_{\text{odd}}}$ and $\mathbf{1}_{B_{\text{odd}}}$ are in the kernel. The corrected theorem states the sharper and correct four-indicator result.

Corollary 11.3 (Kernel dimension split). *The four parity-block indicators account for 4 of the 10 dimensions of $\ker H_{\text{wind}}(8)$. The remaining 6 dimensions come from wrap-corrected combinations of the same parity blocks.*

12 Finite-volume selection of $L = 8$

Theorem 12.1 (Maximal cancellation selects $L = 8$). *Among even integers $L \geq 6$, the maximum cancellation depth of $H_{\text{wind}}(L)$ on its parity block graph is attained uniquely at $L = 8$, where it equals 10.*

Proof. By Theorem 9.1, the depth is 4 for $L \equiv 2 \pmod{4}$, 7 for $L \equiv 0 \pmod{4}$ with $L \geq 12$, and 10 for $L = 8$. The maximum is 10, attained uniquely at $L = 8$. The case $L = 4$ is excluded because the substrate has only $2(L - 1) = 6$ modes, insufficient to host one Standard Model generation. $\square$

Corollary 12.2 (Finite-volume neutrality). *At $L = 8$ the framework coefficients $\{c = 5/9, (d - 1)/2 = 3/2, \dim \ker H_{\text{wind}} = 10\}$ are recovered. At every other even $L \geq 6$, at least one of these fails.*

Remark 12.3. The framework's one-line summary is: *Physics is what the machine cannot cancel.* At $L = 8$ the machine cancels the most while still leaving a Standard Model behind.

13 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script constructs:

1. the 14-mode coordinates;
2. the coordinate vectors n_1, n_2 ;
3. the winding matrix M and phase ϕ ;
4. the winding generator $H_{\text{wind}} = i \sin \phi$;
5. the kernel basis U_{ker} by SVD of H_{wind} ;
6. the projector $P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T$;
7. the permutation matrices P_F and P_{rs} ;
8. the parity block graph $\mathcal{G}_L$ for even L ;
9. the family blocks H_A and H_B ;
10. the four parity-block indicators at $L = 8$.

The script verifies:

1. M_{ij} even integer for all i, j ;
2. H_{wind} Hermitian, entries in $\mathbb{Q}(\sqrt{2})$;
3. $\text{tr}(H_{\text{wind}}) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^4) = 2176$;
4. characteristic polynomial $\lambda^{14} - 40\lambda^{12} + 256\lambda^{10}$;
5. spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$;
6. class partition $\{A, B, C\}$ and its unique orbit-swap form;
7. Kernel-Diagonal Theorem values $\{1/2, 3/4\}$;
8. F -cycle structure and F -invariance of P_{ker} ;
9. kernel- T decomposition: 6, 1, 3;
10. cancellation depth for even $L \in \{4, 6, \dots, 40\}$;
11. family-block eigenvalues $\pm L/2$;
12. four kernel constants at $L = 8$;
13. rank formula for even L .

14 Conclusion

This paper has established the winding generator on the 14-mode multiplet at $\lambda = 4$ of the periodic 8×8 torus, together with its spectrum, its kernel projector, and its finite-volume selection of the substrate size. The generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8} M\right), \quad M = n_1 n_2^T - n_2 n_1^T. \quad (41)$$

It is the unique Hermitian operator on the 14-mode multiplet determined by the two coordinate vectors of the mode labels. Its characteristic polynomial is

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32), \quad (42)$$

derived from the trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ together with a rank-4 parity argument. Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (43)$$

The 14 modes partition into three winding classes A, B, C of sizes $(3, 7, 4)$, uniquely determined by the D_4 orbit partition together with the single two-mode swap $\{t, \text{dark}\}$. The kernel projector P_{ker} of H_{wind} carries the framework's three leading coefficients on its diagonal. Its two-value structure $\{1/2, 3/4\}$ is a consequence of F -invariance of the kernel. The kernel- T decomposition is $(6, 1, 3)$, with the unique C -component $(e_Z - e_H)/\sqrt{2}$. The generator anticommutes with the chiral reflection rs of D_4 .

The finite-volume analysis shows that the cancellation depth and rank of $H_{\text{wind}}(L)$ take a two-case form governed by $L \bmod 4$, that family-block eigenvalues are exactly $\pm L/2$, and that $L = 8$ is the unique even size at which the framework coefficients are recovered. The substrate is not chosen; it is forced by the cancellation ledger.

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